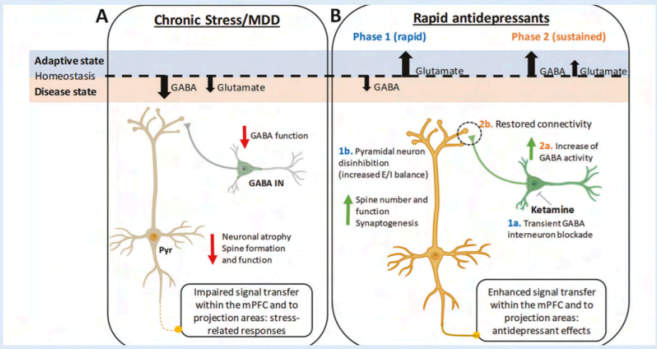
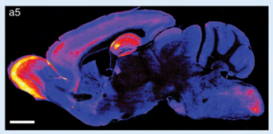
Rapid and Selective Antidepressants: a5-GABAAR Positive Allosteric Modulation Promotes Adaptive Behavior and Cortical VGAT upregulation in mice

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BACKGROUND





The a5-containing GABAA receptor emerges as potential target for its restricted cortico-hippocampal expression [3].

Ketamine rapidly reverses behavioral and biological deficits in Major Depressive Disorder and chronic stress, but its widespread activity confers psychotomimetic effects and abuse potential [1].

The negative allosteric modulator (NAM) of the a5-GABAAR, Basmisanil, induces similar antidepressant-like effects to ketamine, but not conditioned place preference. [2]

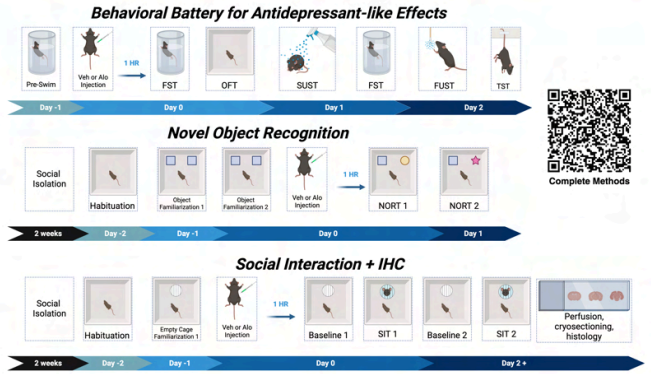


- Ketamine-like antidepressant-like behavioral effects, but no CPP [2]
- ↑ adaptive behavior [2]
- immediate glutamate surge then gradual increase of GABA activity [2]

Paradoxically, GABAAR positive allosteric modulators (PAMs) also reported to produce antidepressant-like effects [4].

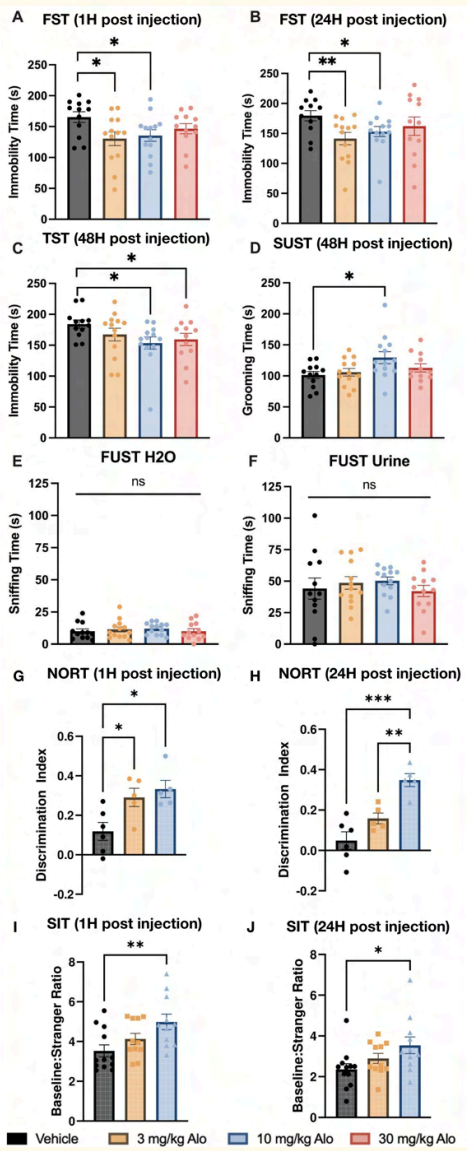
Does bidirectional a5-GABAAR modulation lead to convergent behavioral outcomes? I tested the effects of a5-GABAAR PAM, Alogabat, on mouse behavior and mPFC synaptic markers to explore this question.

METHODS



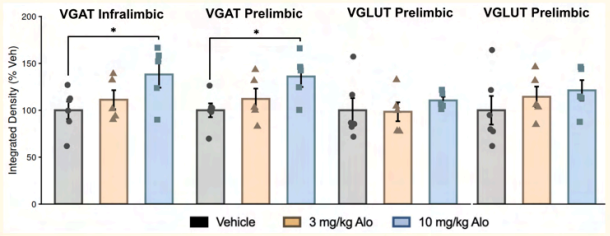
Complete Methods

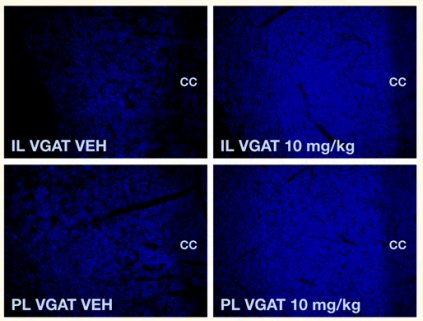
ADAPTIVE BEHAVIORS



Alogabat promotes rapid and sustained antidepressant-like effects in naive male mice (A-F). (A) 3 and 10 mg/kg Alo decreased FST immobility time ($p = 0.022$; $p = 0.029$) 1H and (B) 24H after admin ($p = 0.009$; $p = 0.036$). (C) 10 mg/kg and 30 mg/kg Alo decreased TST immobility time ($p = 0.019$; 0.049). (D) Alo 10 mg/kg increased SUST grooming time ($p = 0.022$). (E, F) Alo exerted no significant effect on sniffing time in FUST. n = 12-13/group.
Alogabat improves discrimination between familiar and novel object in single housed mice (G,H). (A) 3 mg/kg and 10 mg/kg Alo increased discrimination index 1H ($p = 0.048$, $p = 0.014$) after admin. (B) 10 mg/kg Alo increased discrimination index 24H after admin ($p = 0.0002$, $p = 0.007$). n = 5-6/group.
Alogabat increases social behavior in single-housed mice. (I, J) 10 mg/kg Alo increased time spent interacting with a stranger versus an empty cage both 1H ($p=0.009$) and (D) 24H after admin ($p = 0.036$). n = 11-12 mice/group.

CORTICAL GABAergic EFFECTS





Alogabat increases inhibitory marker density in the medial prefrontal cortex of male mice. 10 mg/kg Alo significantly increased VGAT integrated density in the Infralimbic Cortex ($p = 0.046$) and (I) the Prelimbic Cortex ($p = 0.037$) in comparison with vehicle. No significant effect of Alo was observed for VGLUT integrated density. Bars represent mean ± SEM expressed as a percentage of the vehicle group; points represent individual animals (n = 4-6/group).
Representative VGAT and VGLUT1 immunofluorescence images: Magnification 20×

DISCUSSION, FUTURE DIRECTIONS

Alogabat induced varied rapid and sustained antidepressant-like effects, similar to those of Basmisanil, in assays measuring adaptive behaviors (active coping, self care, cognititon, sociality)

- No effect on FUST, diverging from how Basmisanil and Ketamine promote motivated behavior
- Different doses exert most salient effects at different timepoints
- SIT and NORT are more selectively dependent on cortico-hippocampal circuits than traditional antidepressant measures, consistent with hypothesis that a5-GABAAR modulation exerts therapeutic effects via normalization of E/I in mPFC and projections [2].

Alogabat selectively increased levels of VGAT, but not VGLUT1, in the mPFC 24H post-admin.

- Enhancement of GABAergic synaptic proteins mPFC regions consistent with receptor expression and effects on mPFC-dependent behaviors
  - Basmisanil increases VGAT in PL and VGLUT1 in both PL and IL subregions of the mPFC 24H post-admin [2]
  - Ketamine ameliorates stress-induced deficits in VGLUT1, VGAT in the mPFC 24H post-admin [2,4]

Extant Questions/Future Directions:

- Where do the pharmacological effects of Alogabat, Basmisanil, and Ketamine converge and diverge to induce their behavioral effects?
- Why do we observe behavioral effects within 1H of Alogabat administration when PAMs lack transient gluatamatergic "burst" disinhibition?
  - C-fos or fiber photometry for better temporal resolution
- Futher behavioral investigation: effects on anhedonia, replication in chronically stressed mice, and across sexes

ACKNOWLEDGEMENTS

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Citations

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